

Analytic Matrix Elements for Gaussian Basis Functions

Complete Derivation of Overlap, Kinetic, and Potential Integrals
for the Rothe Time-Stepping Method

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1 Introduction and Notation

This document provides a complete derivation of all matrix elements used in the JAX-based Rothe solver for the time-dependent Schrödinger equation. We work in D spatial dimensions with n non-orthogonal Gaussian basis functions.

1.1 Gaussian Basis Functions

Definition 1 (Generalized Gaussian). Each basis function $g_i(\mathbf{x})$ for $i = 1, \dots, n$ is a product of one-dimensional Gaussians:

$$g_i(\mathbf{x}) = \prod_{d=1}^D \phi_i^{(d)}(x_d), \quad \phi_i^{(d)}(x) = \exp \left[-\alpha_i^{(d)}(x - \mu_i^{(d)})^2 + ip_i^{(d)}x \right] \quad (1)$$

where the complex width parameter is

$$\alpha_i^{(d)} = (a_i^{(d)})^2 + ib_i^{(d)}, \quad a_i^{(d)} > 0. \quad (2)$$

The parameters stored for each Gaussian are:

- $a_i^{(d)} > 0$: width parameter (real, stored as a so that $\text{Re}(\alpha) = a^2 > 0$)
- $b_i^{(d)} \in \mathbb{R}$: quadratic phase (chirp)
- $\mu_i^{(d)} \in \mathbb{R}$: center position
- $p_i^{(d)} \in \mathbb{R}$: momentum (linear phase)

These are stored in an array of shape $(n, D, 4)$ with order (a, b, μ, p) .

1.2 Wavefunction Ansatz

The wavefunction is a linear combination:

$$\Psi(\mathbf{x}, t) = \sum_{i=1}^n c_i(t) g_i(\mathbf{x}; \boldsymbol{\theta}_i(t)), \quad (3)$$

where $c_i \in \mathbb{C}$ are complex coefficients and $\boldsymbol{\theta}_i = (a_i, b_i, \mu_i, p_i)$ are the Gaussian parameters.

2 Overlap Matrix S

2.1 One-Dimensional Gaussian Overlap

Consider the overlap of two 1D Gaussians:

$$\langle \phi_i | \phi_j \rangle = \int_{-\infty}^{\infty} \phi_i^*(x) \phi_j(x) dx. \quad (4)$$

Expanding the integrand:

$$\phi_i^*(x) \phi_j(x) = \exp \left[-\alpha_i^*(x - \mu_i)^2 - ip_i x \right] \cdot \exp \left[-\alpha_j(x - \mu_j)^2 + ip_j x \right] \quad (5)$$

Expanding the quadratics:

$$-\alpha_i^*(x - \mu_i)^2 - \alpha_j(x - \mu_j)^2 = -(\alpha_i^* + \alpha_j)x^2 + 2(\alpha_i^*\mu_i + \alpha_j\mu_j)x - \alpha_i^*\mu_i^2 - \alpha_j\mu_j^2 \quad (6)$$

Adding the linear phase terms $-ip_ix + ip_jx = i(p_j - p_i)x$:

$$\phi_i^*(x)\phi_j(x) = \exp \left[- \underbrace{(\alpha_i^* + \alpha_j)}_{\Gamma} x^2 + \underbrace{2(\alpha_i^* \mu_i + \alpha_j \mu_j) + i(p_j - p_i)}_{\text{linear coefficient}} x + \underbrace{(-\alpha_i^* \mu_i^2 - \alpha_j \mu_j^2)}_{\text{constant}} \right] \quad (7)$$

Define $A = \Gamma$, $B = 2(\alpha_i^* \mu_i + \alpha_j \mu_j) + i(p_j - p_i)$, $C = -\alpha_i^* \mu_i^2 - \alpha_j \mu_j^2$, so:

$$\phi_i^*(x)\phi_j(x) = \exp [-Ax^2 + Bx + C] \quad (8)$$

Completing the square: $-Ax^2 + Bx = -A \left(x - \frac{B}{2A}\right)^2 + \frac{B^2}{4A}$, giving:

$$\int_{-\infty}^{\infty} \phi_i^*(x)\phi_j(x) dx = \exp \left[C + \frac{B^2}{4A} \right] \cdot \sqrt{\frac{\pi}{A}} = \exp \left[C + \frac{B^2}{4A} - \frac{1}{2} \ln A \right] \cdot \sqrt{\pi} \quad (9)$$

The $\sqrt{\pi}$ factors are absorbed into normalization.

Definition 2 (Pair Intermediates). For a pair of Gaussians (i, j) in dimension d , define:

$$\Gamma_{ij}^{(d)} = \alpha_i^{(d)*} + \alpha_j^{(d)} \quad (10)$$

$$\tilde{\mu}_i^{(d)} = \mu_i^{(d)} - \frac{ip_i^{(d)}}{2\alpha_i^{(d)*}} \quad (11)$$

$$\tilde{\mu}_j^{(d)} = \mu_j^{(d)} + \frac{ip_j^{(d)}}{2\alpha_j^{(d)}} \quad (12)$$

$$\tilde{y}_{ij}^{(d)} = \tilde{\mu}_i^{(d)} - \tilde{\mu}_j^{(d)} \quad (13)$$

$$\tilde{k}_{ij}^{(d)} = \frac{\alpha_i^{(d)*} \alpha_j^{(d)}}{\Gamma_{ij}^{(d)}} \quad (14)$$

The exponent contribution from dimension d (excluding prefactors) is:

$$E_{ij}^{(d)} = -\frac{(p_i^{(d)})^2}{4\alpha_i^{(d)*}} - \frac{(p_j^{(d)})^2}{4\alpha_j^{(d)}} - \tilde{k}_{ij}^{(d)} (\tilde{y}_{ij}^{(d)})^2. \quad (15)$$

2.2 Full Overlap Matrix

Overlap Matrix S

$$S_{ij} = \langle g_i | g_j \rangle = \exp \left[\sum_{d=1}^D \left(E_{ij}^{(d)} - \frac{1}{2} \ln \Gamma_{ij}^{(d)} \right) - \frac{1}{2} \sum_{d=1}^D \ln \left(2\alpha_i^{(d)*} \right) - \frac{1}{2} \sum_{d=1}^D \ln \left(2\alpha_j^{(d)} \right) \right] \quad (16)$$

The last two terms are the diagonal normalization factors ensuring $S_{ii} = 1$ for all parameter values.

Implementation note: In the code, the overlap is computed as:

```
expo_core = sum_d(exponent_contrib - tilde_y^2 * tilde_k)
expo_pref = -0.5 * sum_d(log(Gamma))
diag_expo = -0.5 * sum_d(log(2 * a^2))
S = exp(expo_core + expo_pref - 0.5*(diag_expo_i + diag_expo_j))
```

3 Kinetic Energy Matrix T

The kinetic energy operator is:

$$\hat{T} = -\frac{1}{2} \sum_{d=1}^D \frac{\partial^2}{\partial x_d^2}, \quad (17)$$

where we use units with $\hbar = m = 1$.

3.1 Shifted Gaussian Identity

The key identity for computing kinetic matrix elements is that integrals of polynomials times Gaussians reduce to moments of a Gaussian distribution.

Shifted Gaussian Identity

For a Gaussian overlap $\langle \phi_i | \phi_j \rangle$ with combined width $\Gamma = \alpha_i^* + \alpha_j$, define the effective center:

$$\bar{x} = \frac{\alpha_i^* \tilde{\mu}_i + \alpha_j \tilde{\mu}_j}{\Gamma} \quad (18)$$

Then for any polynomial $P(x)$:

$$\langle \phi_i | P(x) | \phi_j \rangle = \langle \phi_i | \phi_j \rangle \cdot \langle P(x) \rangle_{\bar{x}, \sigma^2} \quad (19)$$

where $\langle \cdot \rangle_{\bar{x}, \sigma^2}$ denotes expectation with respect to a Gaussian centered at $\bar{x}$ with variance $\sigma^2 = 1/(2\Gamma)$.

Specifically, the key moments are:

$$\langle 1 \rangle = 1 \quad (20)$$

$$\langle x - \mu_j \rangle = \bar{x} - \mu_j = -\frac{\alpha_i^*}{\Gamma} (\tilde{\mu}_i - \tilde{\mu}_j) = -\frac{\alpha_i^*}{\Gamma} \tilde{y} \quad (21)$$

$$\langle (x - \mu_j)^2 \rangle = \frac{1}{2\Gamma} + \left(\frac{\alpha_i^*}{\Gamma} \right)^2 \tilde{y}^2 \quad (22)$$

3.2 Action of $-\frac{1}{2}\partial_x^2$ on a Gaussian

For a single dimension, the second derivative of $\phi_j(x) = e^{-\alpha_j(x-\mu_j)^2+ip_jx}$ is:

$$\partial_x \phi_j = [-2\alpha_j(x - \mu_j) + ip_j] \phi_j \quad (23)$$

$$\partial_x^2 \phi_j = [-2\alpha_j + 4\alpha_j^2(x - \mu_j)^2 - 4i\alpha_j p_j(x - \mu_j) - p_j^2] \phi_j \quad (24)$$

Therefore:

$$-\frac{1}{2}\partial_x^2 \phi_j = [\alpha_j - 2\alpha_j^2(x - \mu_j)^2 + 2i\alpha_j p_j(x - \mu_j) + \frac{1}{2}p_j^2] \phi_j \quad (25)$$

3.3 Matrix Element Calculation

Taking $\langle \phi_i | (-\frac{1}{2}\partial_x^2) | \phi_j \rangle$ and using the shifted Gaussian identity:

$$T_{ij}^{(d)} = S_{ij}^{(d)} [\alpha_j - 2\alpha_j^2 \langle (x - \mu_j)^2 \rangle + 2i\alpha_j p_j \langle x - \mu_j \rangle + \frac{1}{2}p_j^2] \quad (26)$$

Substituting the moments and simplifying (using $\tilde{k} = \alpha_i^* \alpha_j / \Gamma$ and the definition of $\tilde{y}$), all terms involving p_j cancel with terms from the exponent, yielding:

Kinetic Matrix T

$$T_{ij} = S_{ij} \cdot \sum_{d=1}^D \left[\tilde{k}_{ij}^{(d)} - 2(\tilde{k}_{ij}^{(d)})^2 (\tilde{y}_{ij}^{(d)})^2 \right] \quad (27)$$

Define the per-dimension kinetic factor:

$$F_{ij}^{(d)} = \tilde{k}_{ij}^{(d)} - 2(\tilde{k}_{ij}^{(d)})^2 (\tilde{y}_{ij}^{(d)})^2. \quad (28)$$

Then $T_{ij} = S_{ij} \sum_d F_{ij}^{(d)}$.

Remark 1. The simplification works because $\tilde{y}$ already encodes the momentum shifts through $\tilde{\mu}_i$ and $\tilde{\mu}_j$. The final formula depends only on $\tilde{k}$ and $\tilde{y}$, which are the natural "reduced" variables for the Gaussian pair.

4 Kinetic Squared Matrix T^2

The squared kinetic operator is:

$$\hat{T}^2 = \frac{1}{4} \left(\sum_{d=1}^D \partial_d^2 \right)^2 = \frac{1}{4} \sum_{d,d'} \partial_d^2 \partial_{d'}^2. \quad (29)$$

This splits into diagonal terms ($d = d'$) and off-diagonal terms ($d \neq d'$).

4.1 Fourth Derivative in One Dimension

For the diagonal piece, we need $\langle \phi_i | \partial_x^4 | \phi_j \rangle$. Applying ∂_x^2 twice to ϕ_j :

$$\begin{aligned} \partial_x^4 \phi_j = & \left[12\alpha_j^2 - 48\alpha_j^3(x - \mu_j)^2 + 16\alpha_j^4(x - \mu_j)^4 \right. \\ & \left. + (\text{terms with } p_j) \right] \phi_j \end{aligned} \quad (30)$$

Taking the matrix element and using the shifted Gaussian identity for moments up to order 4:

$$\langle (x - \mu_j)^4 \rangle = 3\sigma^4 + 6\sigma^2(\bar{x} - \mu_j)^2 + (\bar{x} - \mu_j)^4 \quad (31)$$

where $\sigma^2 = 1/(2\Gamma)$.

After substitution and simplification (momentum terms cancel as before):

Definition 3 (Single-Dimension T^2 Factor).

$$G_{ij}^{(d)} = 4(\tilde{k}_{ij}^{(d)})^4 (\tilde{y}_{ij}^{(d)})^4 - 12(\tilde{k}_{ij}^{(d)})^3 (\tilde{y}_{ij}^{(d)})^2 + 3(\tilde{k}_{ij}^{(d)})^2 \quad (32)$$

This corresponds to $\frac{1}{4} \langle \phi_i | \partial_d^4 | \phi_j \rangle / S_{ij}$.

4.2 Cross-Dimension Contribution

For $d \neq d'$, the operators act on different coordinates and the wavefunction factorizes:

$$\langle g_i | \partial_d^2 \partial_{d'}^2 | g_j \rangle = \langle \phi_i^{(d)} | \partial_d^2 | \phi_j^{(d)} \rangle \cdot \langle \phi_i^{(d')} | \partial_{d'}^2 | \phi_j^{(d')} \rangle \cdot \prod_{d'' \neq d, d'} S_{ij}^{(d'')} \quad (33)$$

This gives $S_{ij} \cdot F^{(d)} \cdot F^{(d')}$ where $F^{(d)}$ is the kinetic factor from before.

4.3 Combining All Terms

The sum over all pairs (d, d') :

$$\sum_{d, d'} F^{(d)} F^{(d')} = \left(\sum_d F^{(d)} \right)^2 \quad (34)$$

But the diagonal terms $d = d'$ contribute $G^{(d)}$, not $(F^{(d)})^2$, so we must correct:

$$\sum_{d \neq d'} F^{(d)} F^{(d')} = \left(\sum_d F^{(d)} \right)^2 - \sum_d (F^{(d)})^2 \quad (35)$$

Kinetic Squared Matrix T^2

$$T_{ij}^2 = S_{ij} \cdot \left[\sum_{d=1}^D G_{ij}^{(d)} + \left(\sum_{d=1}^D F_{ij}^{(d)} \right)^2 - \sum_{d=1}^D (F_{ij}^{(d)})^2 \right] \quad (36)$$

Remark 2. The three terms correspond to:

1. $\sum_d G^{(d)}$: diagonal fourth-derivative contributions
2. $(\sum_d F^{(d)})^2$: all pairs of second derivatives
3. $-\sum_d (F^{(d)})^2$: subtract overcounted diagonal pairs (replaced by G)

5 Polynomial Moments

For polynomial potentials, we need moments of the form $\langle g_i | x_d^k | g_j \rangle$.

5.1 Generating Moments

Definition 4 (Moment Center and Variance). For a pair (i, j) in dimension d :

$$P_{ij}^{(d)} = \frac{\alpha_i^{(d)*} \tilde{\mu}_i^{(d)} + \alpha_j^{(d)} \tilde{\mu}_j^{(d)}}{\Gamma_{ij}^{(d)}} \quad (\text{effective center}) \quad (37)$$

$$R_{ij}^{(d)} = \frac{1}{2\Gamma_{ij}^{(d)}} \quad (\text{effective variance}) \quad (38)$$

Moment Recurrence

The normalized moments $M_k^{(d)} = \langle g_i | x_d^k | g_j \rangle / S_{ij}$ satisfy:

$$M_0^{(d)} = 1 \quad (39)$$

$$M_1^{(d)} = P_{ij}^{(d)} \quad (40)$$

$$M_{k+1}^{(d)} = P_{ij}^{(d)} M_k^{(d)} + k R_{ij}^{(d)} M_{k-1}^{(d)} \quad (41)$$

This is the Hermite polynomial recurrence for moments of a Gaussian distribution.

5.2 Multi-Dimensional Polynomial Expectation

For a monomial $\mathbf{x}^{\mathbf{n}} = \prod_{d=1}^D x_d^{n_d}$:

$$\langle g_i | \mathbf{x}^{\mathbf{n}} | g_j \rangle = S_{ij} \prod_{d=1}^D M_{n_d}^{(d)}. \quad (42)$$

6 Polynomial Potential V_{poly}

A polynomial potential is:

$$V_{\text{poly}}(\mathbf{x}) = \sum_{\mathbf{n}} v_{\mathbf{n}} \mathbf{x}^{\mathbf{n}}, \quad (43)$$

where the sum is over a finite set of multi-indices $\mathbf{n} = (n_1, \dots, n_D)$.

Polynomial Potential Matrix

$$(V_{\text{poly}})_{ij} = \sum_{\mathbf{n}} v_{\mathbf{n}} S_{ij} \prod_{d=1}^D M_{n_d}^{(d)} \quad (44)$$

7 Gaussian Potential V_{gauss}

The Gaussian potential is a sum of Gaussian terms:

$$V_{\text{gauss}}(\mathbf{x}) = \sum_{k=1}^m c_k \prod_{d=1}^D \exp \left[-\alpha_k^{(d)} (x_d - \mu_k^{(d)})^2 \right], \quad (45)$$

where c_k are coefficients and $\alpha_k^{(d)} > 0$ (real, purely Gaussian width).

7.1 Convolution with Basis Functions

The matrix element $\langle g_i | V_k | g_j \rangle$ for a single Gaussian potential term involves combining three Gaussians. The key insight is that the product of two Gaussians is again a Gaussian.

Definition 5 (Combined Gaussian Parameters). For basis pair (i, j) and potential k :

$$A_{ij}^{(d)} = \alpha_i^{(d)*} + \alpha_j^{(d)} \quad (\text{from basis overlap}) \quad (46)$$

$$B_{ij}^{(d)} = 2(\alpha_i^{(d)*} \mu_i^{(d)} + \alpha_j^{(d)} \mu_j^{(d)}) + i(p_j^{(d)} - p_i^{(d)}) \quad (47)$$

$$C_{ij}^{(d)} = -\alpha_i^{(d)*} (\mu_i^{(d)})^2 - \alpha_j^{(d)} (\mu_j^{(d)})^2 + i(\mu_i^{(d)} p_i^{(d)} - \mu_j^{(d)} p_j^{(d)}) \quad (48)$$

When adding potential k :

$$\tilde{A}^{(d)} = A_{ij}^{(d)} + \alpha_k^{(d)} \quad (49)$$

$$\tilde{B}^{(d)} = B_{ij}^{(d)} + 2\alpha_k^{(d)} \mu_k^{(d)} \quad (50)$$

$$\tilde{C}^{(d)} = C_{ij}^{(d)} - \alpha_k^{(d)} (\mu_k^{(d)})^2 \quad (51)$$

7.2 Gaussian Potential Matrix Element

Completing the square:

$$\text{center} = \frac{\tilde{B}^{(d)}}{2\tilde{A}^{(d)}}, \quad \text{constant} = \tilde{C}^{(d)} + \frac{(\tilde{B}^{(d)})^2}{4\tilde{A}^{(d)}}. \quad (52)$$

Gaussian Potential Matrix

$$(V_{\text{gauss}})_{ij} = \sum_{k=1}^m c_k \exp \left[\sum_{d=1}^D \left(\text{constant}_k^{(d)} - \frac{1}{2} \ln \tilde{A}_k^{(d)} - \frac{1}{2} E_{\text{sub}}^{(d)} \right) \right] \quad (53)$$

where $E_{\text{sub}}^{(d)}$ is the normalization subtraction term (diagonal elements of basis overlap).

8 Potential Squared V^2

The squared potential involves:

$$\langle g_i | V^2 | g_j \rangle = \langle g_i | V_{\text{poly}}^2 + V_{\text{gauss}}^2 + 2V_{\text{poly}}V_{\text{gauss}} | g_j \rangle. \quad (54)$$

8.1 Polynomial Squared

For V_{poly}^2 , we simply compute the squared polynomial by multiplying out all terms and collecting coefficients for each monomial $\mathbf{x}^{\mathbf{m}}$.

If $V_{\text{poly}} = \sum_{\mathbf{n}} v_{\mathbf{n}} \mathbf{x}^{\mathbf{n}}$, then:

$$V_{\text{poly}}^2 = \sum_{\mathbf{n}, \mathbf{n}'} v_{\mathbf{n}} v_{\mathbf{n}'} \mathbf{x}^{\mathbf{n}+\mathbf{n}'}. \quad (55)$$

8.2 Gaussian Squared

For two Gaussian potential terms k and k' , their product is:

$$V_k V_{k'} = c_k c_{k'} \prod_{d=1}^D \exp \left[-(\alpha_k^{(d)} + \alpha_{k'}^{(d)})(x_d - \mu_{kk'}^{(d)})^2 + \text{const} \right], \quad (56)$$

which is again a Gaussian. The squared Gaussian potential is precomputed by forming all m^2 product terms.

8.3 Polynomial-Gaussian Cross Terms

For $V_{\text{poly}}V_{\text{gauss}}$, each Gaussian potential modifies the effective center and variance for the moment calculation:

$$P_k^{(d)} = \frac{\tilde{B}_k^{(d)}}{2\tilde{A}_k^{(d)}} \quad (\text{new center}) \quad (57)$$

$$R_k^{(d)} = \frac{1}{2\tilde{A}_k^{(d)}} \quad (\text{new variance}) \quad (58)$$

The moments are recomputed with these modified parameters, then contracted with the polynomial coefficients.

9 Kinetic-Potential Cross Terms $TV + VT$

The Hamiltonian squared contains:

$$H^2 = T^2 + V^2 + TV + VT. \quad (59)$$

Since T and V generally don't commute, we need both TV and VT .

9.1 Hermiticity Relation

Since both T and V are Hermitian:

$$(VT)_{ij}^* = \langle g_j | VT | g_i \rangle^* = \langle g_i | TV | g_j \rangle = (TV)_{ij}. \quad (60)$$

Thus: $\boxed{TV = (VT)^\dagger}$.

We only need to compute VT and obtain TV by conjugate transpose.

9.2 Polynomial VT

Acting with $T = -\frac{1}{2}\nabla^2$ on $|g_j\rangle$ and taking the matrix element with V_{poly} :

Polynomial VT Matrix

For monomial $\mathbf{x}^{\mathbf{n}}$ with coefficient $v_{\mathbf{n}}$, the contribution to $(VT)_{ij}$ involves:

$$\text{base}^{(d)} = -2(\tilde{k}^{(d)})^2(\tilde{y}^{(d)})^2 + \tilde{k}^{(d)} \quad (61)$$

and correction terms from the polynomial:

$$\text{vt_coeff}^{(d)} = \text{base}^{(d)} - 2n_d \tilde{k}^{(d)} \tilde{y}^{(d)} \beta_j^{(d)} \frac{M_{n_d-1}}{M_{n_d}} - \frac{1}{2} n_d (n_d - 1) (\beta_j^{(d)})^2 \frac{M_{n_d-2}}{M_{n_d}} \quad (62)$$

where $\beta_j^{(d)} = \alpha_j^{(d)} / \Gamma_{ij}^{(d)}$.

Then:

$$(VT)_{ij} = \sum_{\mathbf{n}} v_{\mathbf{n}} S_{ij} \prod_{d=1}^D M_{n_d}^{(d)} \cdot \sum_{d=1}^D \text{vt_coeff}^{(d)} \quad (63)$$

9.3 Gaussian VT

For Gaussian potentials, we compute expectation values of the Laplacian in the presence of the Gaussian weight.

Definition 6 (Gaussian-Weighted Moments). With potential k present:

$$\langle (x - \mu_k)^2 \rangle_k = R_k + (P_k - \mu_k)^2 \quad (64)$$

$$\langle (x - \mu_j)(x - \mu_k) \rangle_k = R_k + (P_k - \mu_j)(P_k - \mu_k) \quad (65)$$

Gaussian VT Matrix

For each potential term k :

$$\text{term}_1 = -\frac{1}{2} \sum_d [4\alpha_k^2 \langle (x - \mu_k)^2 \rangle_k - 2\alpha_k] \quad (66)$$

$$\text{term}_2 = - \sum_d (-2\alpha_k) \left[(-2\alpha_j) \text{Cov}_{kj}^{(d)} + ip_j \langle x - \mu_k \rangle_k \right] \quad (67)$$

$$\text{term}_3 = - \sum_d [-2\alpha_j + 4\alpha_j^2 \langle (x - \mu_j)^2 \rangle_k - 4i\alpha_j p_j \langle x - \mu_j \rangle_k - p_j^2] \quad (68)$$

where $\text{Cov}_{kj}^{(d)} = R_k + (P_k - \mu_k)(P_k - \mu_j)$.

Then:

$$(VT)_{ij} = \sum_{k=1}^m c_k \cdot \text{prefactor}_k \cdot (\text{term}_1 + \text{term}_2 + \text{term}_3) \quad (69)$$

The full cross term is:

$$TV + VT = VT + VT^\dagger. \quad (70)$$

10 The Rothe Method

10.1 Time-Stepping Formulation

The Rothe method approximates the time evolution by minimizing the residual of the implicit Euler scheme:

$$\left\| \frac{\Psi^{n+1} - \Psi^n}{\Delta t} + iH\Psi^{n+1} \right\|^2 = \min. \quad (71)$$

10.2 Matrix Form

Expanding in the Gaussian basis $\Psi^{n+1} = \sum_j c_j^{n+1} g_j^{n+1}$:

Rothe Error Functional

$$\mathcal{E}[\boldsymbol{\theta}^{n+1}] = \boldsymbol{c}^{n+1\dagger} \left(S + i\Delta t H + \frac{\Delta t^2}{2} H^2 \right) \boldsymbol{c}^{n+1} - 2\text{Re} \left(\boldsymbol{c}^{n+1\dagger} S_{01} \boldsymbol{c}^n \right) + \langle \Psi^n | \Psi^n \rangle \quad (72)$$

where:

- S is the overlap matrix of new basis functions
- H is the Hamiltonian matrix in the new basis
- H^2 is the squared Hamiltonian matrix
- S_{01} is the overlap between old and new bases

10.3 Optimal Coefficients

For fixed Gaussian parameters $\boldsymbol{\theta}^{n+1}$, the optimal coefficients are:

$$\boldsymbol{c}^{n+1} = \left(S + i\Delta t H + \frac{\Delta t^2}{2} H^2 \right)^{-1} S_{01} \boldsymbol{c}^n. \quad (73)$$

The optimization then proceeds over the Gaussian parameters θ^{n+1} .

11 Summary of Intermediates

Symbol	Description	Shape
$\alpha_i^{(d)}$	Complex width: $a_i^2 + ib_i$	(n, D)
$\Gamma_{ij}^{(d)}$	Sum: $\alpha_i^* + \alpha_j$	(n, n, D)
$\tilde{y}_{ij}^{(d)}$	Shifted center difference	(n, n, D)
$\tilde{k}_{ij}^{(d)}$	$\alpha_i^* \alpha_j / \Gamma_{ij}$	(n, n, D)
$P_{ij}^{(d)}$	Effective center for moments	(n, n, D)
$R_{ij}^{(d)}$	Effective variance: $1/(2\Gamma)$	(n, n, D)
$\beta_j^{(d)}$	Ratio: α_j / Γ_{ij}	(n, n, D)
$M_k^{(d)}$	k -th moment (normalized)	$(n, n, D, K_{\max} + 1)$

Table 1: Key intermediate quantities computed once per basis update.

12 Implementation Notes

12.1 Memory Efficiency

- Gaussian potential sums are computed via `lax.scan` to avoid materializing (m, n, n, D) tensors.
- Custom VJP rules provide memory-efficient gradients for the backward pass.
- Moments are precomputed up to $K_{\max}$ determined by the polynomial degree.

12.2 GPU Optimization

- The Gaussian kinetic cross terms are vectorized over the m potential terms.
- Avoid `lax.fori_loop` for small m as it serializes GPU work.
- Use `tensordot`-style contractions rather than explicit loops.

12.3 Numerical Stability

- All computations use `complex128` for accuracy.
- Overlap matrix is computed in log-space to avoid underflow.
- Safe division with ϵ guards for moment ratios.